KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCES

COURSE: SYSTEM VULNERABILITY ASSESSMENT **COURSE CODE: CYS 330**
DATE: 1st AUGUST 2025 **TASK: TEST 2**

Instructions

1. Write your **Student Number** on the **Answer Sheet (Booklet)** provided
2. There are **four (4)** questions in this paper. Marks are allocated at the end of each question. Each question carries **25 Marks.**
3. Total marks for this paper: **100%**
4. You are allowed to have an allowance of **5 minutes** to go through your paper before the beginning of the Test. Note that a good hand-writing is encouraged.
5. Duration: **2 Hour**
6. No scientific and programmed calculators will be allowed in this paper

Question one [25 Marks]

1. An overflow counts for about 23% according to the CVSS of the attack and therefore, requires deep knowledge and understanding in order to easily identify and mitigate it as a vulnerability. This introduces one's and two's compliment numbers in binary number system which the computer is believed to be using.

   (a) Add the following numbers in two's compliment using 8 bits. Determine whether an overflow has occurred. Show all your working **[6 Marks]**
   (i) -47 and 19 ✓
   (j) 98 and 74

   (b) Add the following numbers in one's compliment using 8 bits. Determine whether an overflow has occurred **[6 Marks]**
   (i) 25 and -56 ✓
   (ii) -77 and -77

   (c) Subtract the following numbers in one's compliment using 8 bits. Determine whether an overflow has occurred. **[6 Marks]**
   (i) 66 – (-78)
   (ii) 72 – 25

   (d) Subtract the following numbers in two's compliment using an octet. Determine whether an overflow has occurred **[6 Marks]**
   (i) 70 – 32 ✓
   (ii) 99 – (71) ✓

2. What is an overflow? **[1]** (1)

Question two [25 Marks]

1. Mitigation and Remediation require planning in advance. PERT can be used to visualise these plans and make it easy for IT and Project Managers to follow through.

   (a) Construct a network diagram with the data given below **[5]** (5)(5)

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Activity	Predecessor (s)	Duration
A	-	5
B	-	3
C	A	10
D	B	2
E	C, D	6

(b) Compute the Project duration **[2]**
(c) Determine the critical path **[10]**
(d) Draw a project network for a remediation plan from the following statement **[8]**
E > B, C; F, G > D; H, I > E, F; J > I, G; K > H; B > A;

Question Three [25 Marks]

1. Activities of a remediation plan has been assigned to you to come up with the project network which is scheduled to be presented in a managing meeting.

Activity	Duration
1 – 2	13
1 – 3	12
2 – 4	2
3 – 4	8
2 – 5	15
4 – 5	2

(a) Draw a network diagram for the remediation plan project **[5]**
(b) Determine the critical path and find the Earliest Start (ES), Earliest Finish (EF), Latest Start (LST), and Latest Finish (LF) **[10]**
(c) Calculate the Total Float (TF), Free Float (FF), Independent Float (INT F), and the Interfering Float (INT F) **[10]**

Question four [25 Marks]

The following have been given for your convenient as an IT Manager in charge of all ICT work related project

Activity	Optimistic	Most Likely	Pessimistic
1 – 2	3	6	15
2 – 3	2	5	14
1 – 4	6	12	30
2 – 5	2	5	8
2 – 6	5	11	17
3 – 6	3	6	15
4 – 7	3	9	27
5 – 7	1	4	7
6 – 7	2	5	8

(a) Draw the corresponding network diagram **[5]**
(b) Compute the expected duration for each activity **[5]**
(c) Calculate the expected variance of each activity **[5]**
(d) Determine the expected critical path variance **[5]**
(e) Compute the standard deviation of each activity **[5]**

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